Hostel-Ease: A Web-Based Hostel Management System and Leave Management Using MERN Stack

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Abstract—*The management of hostels is one of the most important administration functions of institutions of learning, which includes accommodation of students, rooms allocation, handling of complaints, and control of leaves. Conventional manual recording systems with registers and paper work are associated with delays, lapses in maintenance of records, lack of transparency and amplified workload. All these complicate the management of the hostel institutions in terms of effective management of operations and maintenance of proper records. Hostel-Ease is an automated web-based hostel management system that will help address such problems and streamline the operations in a centralized digital platform. The system enables students to enroll, log-in, see room information, demand room assignment, leave request, and make a complaint interactively. The administrators are able to handle student records, room availability, and requests via a specialized dashboard. The system is created with MERN stack and runs on Vercel and Render and also incorporates mail service to recover the password and notification, which eventually saves lots of paperwork and enhances transparency and efficiency.*

***Keywords*—**Hostel Management System, Web Application, MERN Stack, Room Allocation, Complaint Management, Leave Management, React.js, Node.js, MongoDB Atlas, Hostel-Ease.

I. INTRODUCTION

Hostel activities management is an extremely significant administrative exercise in learning facilities since students rely on the hostel facilities to accommodate, assign rooms, provide abodes to complaints, and handling leave. These activities are still handled manually by most institutions by use of registers, paper forms or informal communication. Such a conventional system tends to create delays in the processing, non-transparency, and inability to keep accurate records. The more the students are, the more time The bigger the numbers of students, the more time-consuming and ineffective it becomes to the administrators to control the operations of the hostel manually. it takes and ineffective the administrators to manually handle the hostel operations. As web technologies are developed, these administrative processes can be automated in order to increase their efficiency, accuracy, and accessibility.

A digital Hostel Management System is a web-based phenomenon through which students get a centralized and easy to access system whereby they can register and apply to rooms, they can request leave, and raise complaints as they see their position change in real time. All the requests are handled via a dashboard and the administrators can save on the paperwork, better communications and make the running of the hostel effective and well organized.

II. LITERATURE SURVEY

Hostels in learning institutions have been improved through the use of digital technologies to manage them. The hostel management had been maintaining the registers and paperwork to track down the activities like room allocation, complaints management among others. This was a conventional method that was time-wasting, prone to errors which was mostly unproductive particularly when there were a number of students. In order to overcome these issues web-based hostel management systems were initiated. The first systems were simple with different functions to store the information about students and control-rooms, however, they did not offer sophisticated knowledge like real-time update and effective communication.

Modern technologies such as MERN stack have been used to develop more complex and simpler systems. Such systems are characterized by centralized data handling, superior processing speed, and improved interfaces, but some of them have certain shortfalls such as low scalability and lack of user-friendliness. Hostel-Ease proposed system eliminates these shortcomings by offering a single-stop solution to booking rooms, complaints, and leave management and cloud support as well as enhanced communication priorities.

A) Problem Statement

1) *Research Question: What can be done to enhance the hostel management in terms of assigning rooms, managing complaints and leave system through efficient and safe online application?*

2) *Relevance:*

a. Administrative Inefficiency: Work is complicated and slow due to paper work.

b. Record Management Issues: The paper-based records are difficult to store and sustain.

c. Inadequate level of Transparency: Learners cannot follow requests resulting in the issue of communication gap.

III. PROPOSED SYSTEM

The current system is the proposed system, which is known as Hostel-Ease; a web-based hostel management system that will be adopted to automate and streamline hostel operations in learning institutions. The system offers an online platform, which is easy to conduct business between administration and students without necessarily carrying out more manual operations, but enhances transparency.

The system starts its workflow with the user authentication, in which the students and administrators log in safely and view the corresponding dashboards. Once a student logs in, he or she is able to see the available rooms, request to be assigned a room, file a leave request as well as file a complaint. All these requests are saved in the database and directed to the administrator to be processed. Room allocation module enables students to check the availability of rooms and request a room, whereas the administrator verifies and assigns rooms, according to the availability. The complaint management module allows students to give out the problems, which are discussed and addressed by the administrator. The leave management module enables students to request leave and the leave is either approved or denied by the admin. Everything is monitored and reported instantly, enabling students to track the situation with their requests.

The dashboard can also enable the administrator to create reports and to track the work of the hostel in general. It is a system created with the latest web technologies where the front end is developed with React.js and the backend services are created with Node.js and express.js and the database is being managed with MongoDB Atlas. It is hosted through Vercel and Render, and it is scalable and accessible. Email services with Resend features are in place on notifications and password recovery. The suggested system will minimize paperwork, improve communication and provide effective and safe management of hostel activities.

1.User Login

New users (students) are registered and information with the necessary details is given. Upon registration, the user can log in with the help of his or her credentials. The system authenticates the data and allows the access to the dashboard depending on the user roles (student/admin).

2. Dashboard Access

On the log-in of the internet, they are redirected to the user dashboards. The services are availed to the students regarding room booking, complaints and leave requests. With administration privileges there is access to management controls and spying devices.ng devices.

3. Room Allocation Process

Students see the rooms available and make a request of allocating a room. The admin verifies the availability of rooms and allocates the rooms. The system changes the status of rooms and informs the student.

4. Complaint Management Process

Complaints on hostel facilities have been done by the students of the hostels. The complaint is viewed by the admin, and the needed action is performed, as well as the status is updated. In the real time, the students are able to track the resolution.

5. Leave Management Process

Applications are made by students by filling in information required. The request is verified by the admin who then approves/rejects it. The system enhances the leave status and shows it in the student dashboard.

6. Status Tracking System

All the requests (room, complaint, leave) are immediately traced. The students are able to check the progress at any particular time making it transparent.

7.Admin Monitoring and Control

The admin will be in charge of all the activities of the system including the users, rooms, complaints, and leaves requests through a centralized dashboard.

8. Notification System

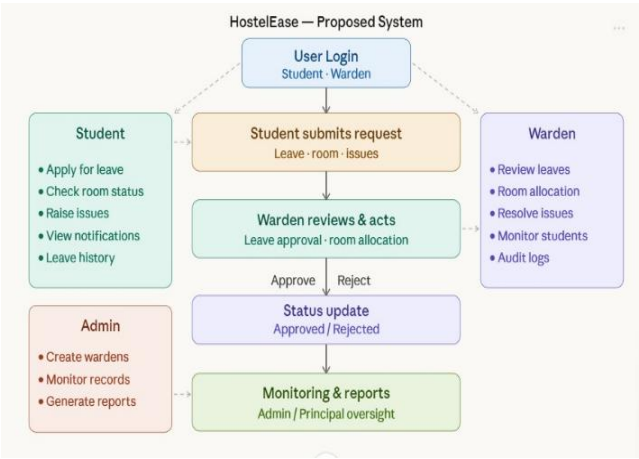
The system also uses the email system to provide email notification (with the help of resend) in the event of password reset and crucial updates, enhancing communication.

9. Data Storing and Management.

MongoDB Atlas is used to store all data in a secure manner. The system is very efficient in data retrieval, changes and scalability.

10: System Deployment and Access.

It is deployed on both frontend (Vercel) and backend (Render) and can be reached in the domain hostel-ease, online, which is secure and, therefore, guarantees performance and availability.



IV. METHODOLOGY

The given methodology explains the functionality of the Hostel-Ease system that is aimed to automate the operations of the hostel in a structured and efficient way. The system also has a particular workflow governing the operations of the user authentication, room allocation, complaint and leave management, gate call monitoring and administration. Everything is centralized to support proper communication and real-time information.

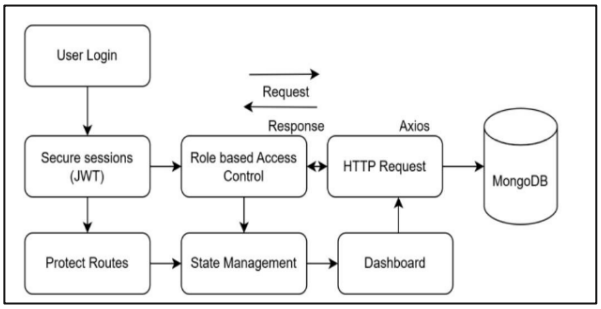


Fig: Client Workflow

The Hostel-Ease system has a user-friendly and secure client-side workflow, which can deliver a secure interface to students and administrators. This starts with user login where the user is checked and an ID is issued as a JWT token which is used to secure sessions. Role-based access control means that users have the capability to use features according to their roles.

Secured paths prevent access by intruders and upon successful user login, the user is redirected to his or her dashboard. Using the React state management, the dashboard depicts real time information like room status, complaints, and leave requests. The HTTP requests are sent by the axios to the backend APIs which are connected to the MongoDB database to store and retrieve data.

The workflow generally has secured authentication, ease of navigation and an effective interaction of the frontend and the backend.

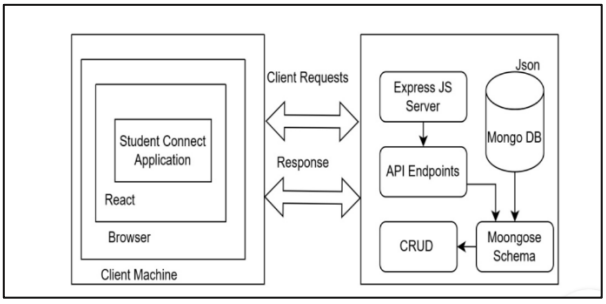


Fig: Backend Architecture

The Hostel-Ease system possesses its back-end which is going to reply to the client requests and process the information, and be able to communicate with the database successfully. When the client (React application) requests the server with the HTTP requests, the process starts. Each API endpoint possesses its operations such as user authentication, room allocation, complaints, leave allocation and tracking of gate entry. The data is validated and business logic put in force by the server, then it is processed. The backend database used is MongoDB database and the schema interface of a backend is the Mongoose which stores data in the JavaScript format. The CRUD (Create, Read, Update, Delete) is the type of operations performed to regulate the user data, rooms, complaints, leave requests, and gate logs. The client then gets the response of the client by the server after the process and this specifies to the user interface. On the whole, the backend workflow will guarantee the security of the data processing process, the efficiency of the data processing and the harmonious relations between the clients and the database.

1. User Authentication

The system begins with the user authentication where students and administrators are required to log-in according to their credentials. The typed in information is verified with the database and controlled access to the system is provided by the token-based authentication. The system will redirect the users to their depending on the roles of the user.

2. User Registration

Users who are new must fill only a little information (name, email and password). The system validates input data and the same is stored in the database in an insecure manner. Once registered, the user is able to log-in and make use of capabilities of the system.

3. Room Allocation Process

Students are able to see what rooms are available by their dashboard and request them to allocate a room. Such requests are saved in the database and sent to the administrator. The admin checks the availability of rooms and allocates rooms. In the case the room is assigned to the student, the system informs the student dashboard about the status of the room with the received information.

4. Complaint Handling

Students may also complain about hostel facilities by giving requisite information. The admin investigates the problem, makes the relevant action, and changes the status of complaints. Students are able to monitor the complaints in real time.

5. Leave Management Process

Students have the option of applying leave by typing in the details of leave duration and the reason. The database adds the leave request which is sent to the administrator.

6.Gate Keeper Monitoring (Entry/Exit Management)

The system will consist of gatekeeper module to track student entry and exit in the hostel. Before a student is allowed to exit or enter the gatekeeper can ensure that the student details have been verified and leave approval status checked. Every movement is captured in the system and there is security and good monitoring of activities of students.

7. Admin Monitoring

The administrator is an important character in the management of the system. Admin is responsible to administer users, rooms, complaints, and leave requests via the dashboard.

8. Email Notification System

The system incorporates the email services of the resend which are used to send notifications on password reset links

as well as updates on user requests. This enhances the communication of the system with users.

9. Data Management

MongoDB Atlas safely stores all the system-related data and information, such as data about users, rooms, complaints, leaves, and gate logs. The database is efficient in data retrieval, scalability and reliability. 10. System Deployment Hostel-Ease application is implemented on contemporary cloud systems. The frontend is hosted on Vercel and backend is deployed on Render. The system is availed on domain hostelease.online, which guarantees usability and functionality.

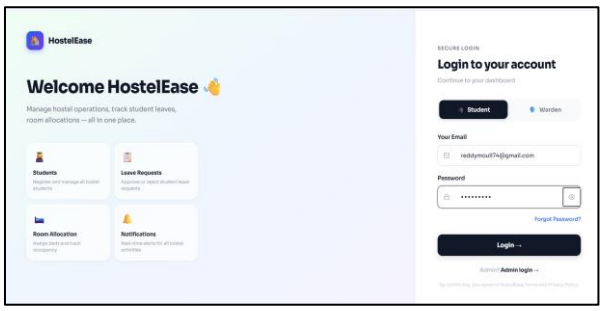


Fig 1. User Login

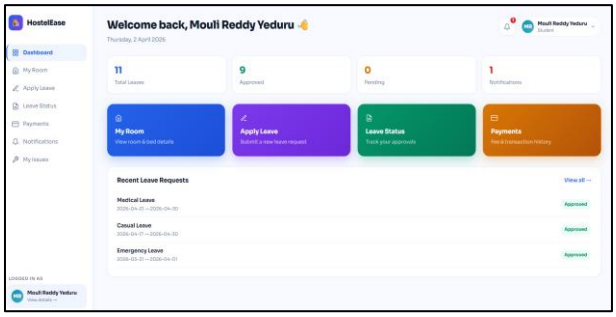


Fig 2. Student Login

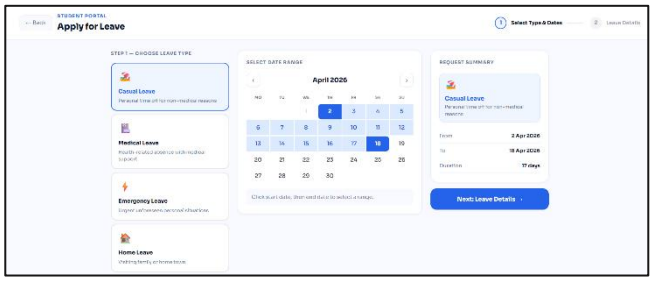


Fig 3. Leave Application

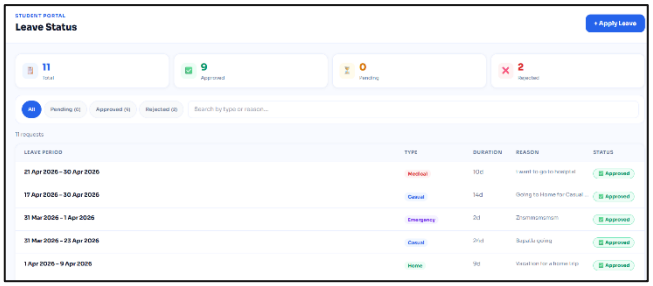


Fig 4. Leave Status

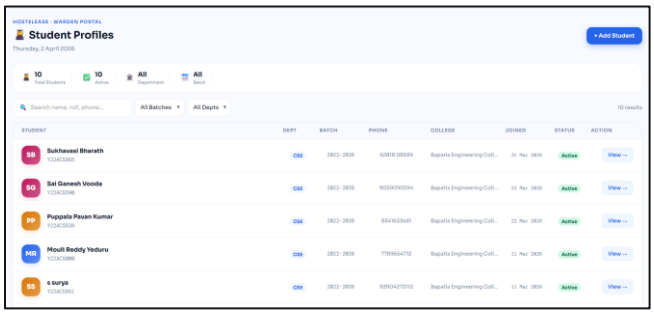


Fig 5. Student List

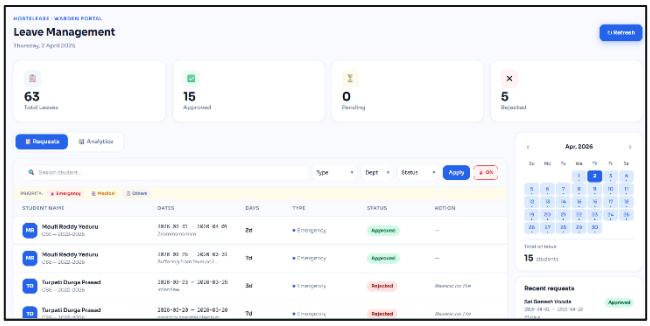


Fig 6. Leave Approvals

V. RESULTS

The Hostel-Ease system was created and deployed successfully based on MERN (MongoDB, Express.js, React.js, and Node.js). The system offers a centralized platform where students request allocation of rooms, apply leave, and also raise complaints whereas the administrator facilitates all the operations by a designed dashboard. The system implementation process led to a substantial decrease in the number of manual labor and paperwork. Activities, which used to be done manually and physical records were used, are now being done digitally hence it is processed faster and with better efficiency. The system provides proper and sensitive record keeping where data can be retrieved and controlled easily.

The role-based access control system is used to ensure that system security is ensured as only people with the required authorization can access particular functionalities. The real-time updates will enable students to monitor the progress of their requests and this will enhance transparency and user satisfaction. The system was also tried in other situations and was identified as reliable, scalable and easy to use. It facilitates effective communication between students and administrators with inbuilt notification. All in all, Hostel-Ease system has proved to be more efficient, there is reduced operational delays, data is handled well and users have a far better experience thus making it a viable and efficient solution to the present management of hostels in learning centers.

VI. CONCLUSION

To sum up, the Hostel-Ease system is effective to overcome the difficulties in the traditional hostel management since it offers centralization and digital approach. The key operations that are automated to automatic systems on the system include room allocation, complaints, leave requests and gate passes, which results in less manual work and eradicates the possibilities of physically error. Using MERN stack the

application enjoys high performance, scalability and responsive user interface. Role-based access is improved by integrating separate student, administrators and gatekeeper dashboards, and increases the overall system efficiency. Updates/notices in real-time are another foundation of strengthening the communication between users and the management and in this context transparency and timely decision-making. Besides, the system facilitates improved data organization and retrieval via a tabled database and thus institutions are able to keep proper records and develop an insight when required. Not only does this decrease in the paperwork save time, but is also a part of the more environmentally friendly methodology. Hostel-Ease is flexible to meet the requirements of various institutions by customizing it. Its modular design enables it to be easily upgraded and added with other functionality in future

VII. FUTURE SCOPE

To make Hostel-Ease system more efficient and user-friendly, it is possible to add new features and upgrade the premium working functions available.

- Cloud Enhancement: The current system is already implemented on cloud solutions like Vercel and Render. Such enhancements that may be introduced in the future are enhanced scalability, security, and automated backup systems.
- Payments-integrations: I can make an addition of an online secure payment system where students can pay their hostel fees online to make it streamlined and easy.
- AI Chatbot Support: AI-assisted chatbot can be implemented to help users with their queries, navigation, and prompt support in the system.
- Biometric/Attendance Integration: integrate the system with biometric or attendance systems to have a more accurate tracking as to student presence and hostel usage.

VIII. REFERENCES

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